# Velkscala Modelling Report - 4 November 2026



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# Mathematical description of Velkscala's Extended Demographic System

The extended system is a constrained demographic reconstruction, with the following order of orchestration in `NWM`:

1. **Age–sex reconstruction** estimates population composition using multinomial logistic models and temporal ensembling.
   - For $t \leq 1750$, graduation operates separately on each pixel: elderly monotonic ceilings, coupled combined-density PAVA with biological sex-ratio bounds, and secondary sex-specific PAVA.
   - For $t > 1750$, graduation operates in ISO3 country bins. A logistic transition over $1750 < t < 1850$ blends coupled combined-density PAVA with coupled Huber–Whittaker graduation. Three-pass Iterative Proportional Fitting redistributes the resulting targets to pixels.
   - HMD, UNWPP, and WorldPop source compositing follows graduation and redistribution.
2. **Migration estimation** constructs empirical-residual or modelled migration, imposes the upper feasibility ceiling $M_{i,t} \leq g_{i,t} + P_{i,t}$, and attempts to remove global imbalance through three population-weighted waterfill iterations over unsaturated cells. A reduced Rogers–Castro schedule supplies sex-allocation weights in the modelled branch.[1]
3. **Birth/death reconciliation**:
   - Preserves designated pre-1950 HMD cells by replacing migration with the local accounting residual;
   - Rebalances migration over unsaturated, non-HMD cells;
   - Applies country-bin bivariate hyperbolic scaling;
   - Closes pixel-level natural change through bivariate hyperbolic reconciliation, with non-negative births and a mortality ceiling on non-HMD cells.

Conservation properties follow from normalisation, constrained adjustment, and local accounting. Regressions estimate spatial variation; they do not independently establish demographic closure.

# Computational domain and temporal conventions

All rasters are assumed aligned on the WGS84 equirectangular $4320 \times 2160$ grid. The nominal temporal domain is:

$$-10000 \leq t \leq 2025.$$

The numeric year $0$ is a technical timeline value, not a claim about historical calendar notation.

Two distinct year grids must be distinguished:

| Construction | Antiquity spacing | Reported total |
|---|---|---|
| Velkscala population-stock composites | 100-year steps from $-10000$ through $0$ | 218 stock rasters |
| Default HYDE-based downstream schedule | 1000-year steps from $-10000$ through $0$ | 128 processing years |

After year $0$, the default downstream schedule uses 100-year steps through 1700, 10-year steps through 1950, and annual steps through 2025. Explicit `options.years` may replace this schedule.

Annualised population-change rasters omit the initial timestep. Consequently, the default flow construction covers 127 intervals, ending at years from $-9000$ through 2025. Population stocks available at additional antiquity dates do not automatically imply corresponding age–sex or flow outputs.

For a processing year with predecessor $t^-$, define:

$$h_t = t - t^- > 0, \qquad g_{i,t} = \frac{P_{i,t} - P_{i,t^-}}{h_t}.$$

The delta raster already stores $g_{i,t}$, not the unannualised difference. It must not be divided by $h_t$ again.

Births, deaths, and migration below are annualised counts. Over a multi-year interval, their accounting interpretation is an interval-average flow consistent with the endpoint stocks, not a reconstruction of every intervening year.

Final PNG flow outputs encode IEEE-754 `binary32` values in RGBA channels. Arithmetic identities established over real numbers therefore need not hold exactly after serialisation.

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# Definition 1. Age–sex reconstruction via MNL with PAVA–Whittaker constraints

## 1.1. Notation and covariates

Let:

- $i \in \Omega$ index raster cells;
- $t$ index processing years;
- $s \in \{f, m\}$ index sex;
- $a \in \mathcal{A} = \{0, \dots, 17\}$ index age bands;
- $b(i)$ denote the ISO3 country bin, with unmapped cells assigned to `GLOBAL`;
- $H, U, W$ denote HMD-, UNWPP-, and WorldPop-derived age–sex sources;
- $P_{i,t} \geq 0$ denote the population-stock raster;
- $N^{s}_{i,t,a} \geq 0$ denote final age–sex population;
- $X_{i,t} \in \mathbb{R}^{35}$ denote the covariate vector.

The age labels are:

$$(0, 1, 5, 10, \dots, 80).$$

Band widths and computational midpoints are:

$$w_a = \begin{cases} 1, & a = 0, \\ 4, & a = 1, \\ 5, & a \geq 2, \end{cases} \qquad x_a = \sum_{c<a} w_c + \frac{w_a}{2}.$$

Thus:

$$x_0 = 0.5, \qquad x_1 = 3, \qquad x_2 = 7.5.$$

The open-ended $80+$ band is assigned computational width $5$. This is an implementation convention, not a finite upper bound on age.

The 35 covariates comprise:

| Group | Count | Raster names |
|---|---|---|
| HYDE land use | 14 | `conv_rangeland`, `cropland`, `grazing`, `irrigated_not_rice`, `irrigated_rice`, `pasture`, `rainfed_not_rice`, `rainfed_rice`, `rangeland`, `rf_norice`, `rf_rice`, `tot_irri`, `tot_rainfed`, `tot_rice` |
| Velkscala population | 4 | `popc_`, `popd_`, `rurc_`, `urbc_` |

| Group | Count | Raster names |
|---|---|---|
| Eoscala economic variables | 9 | `discretionary_income`, `disposable_income`, `gdp_nominal`, `gdp_pc`, `gdp_ppp`, `gdp_ppp_pc`, `gini`, `net_income`, `net_wealth` |
| Annualised changes | 8 | `delta_gdp_nominal`, `delta_gdp_pc`, `delta_gdp_ppp`, `delta_gdp_ppp_pc`, `delta_popd_`, `delta_popc_`, `delta_rurc_`, `delta_urbc_` |

Define the covariate year and age–sex clamping population by:

$$\tilde{t} = \min(t, 2023), \qquad \widetilde{X}_{i,t} = X_{i,\tilde{t}}, \qquad P^{A}_{i,t} = P_{i,\tilde{t}}.$$

Migration and vital-event reconciliation instead read $P_{i,t}$ directly, including the generated population rasters for 2024 and 2025. Therefore:

$$P^{A}_{i,t} = P_{i,2023} \quad \text{for } t \in \{2024, 2025\},$$

which need not equal $P_{i,t}$.

The supplied UNWPP age–sex series ends in 2020, as does WorldPop coverage. These sources do not automatically replace modelled outputs in 2021–2025.

For scalars, write:

$$[z]_{+} = \max(0, z), \qquad \operatorname{clip}(z; \ell, u) = \min\{u, \max\{\ell, z\}\}.$$

# 1.2. Training targets and weighted multinomial models

## Source selection and sampling

For available training years, beginning with HMD coverage in 1750, select targets:

$$T^{s}_{i,\tau,a} = \begin{cases} H^{s}_{i,\tau,a}, & \tau < 1950,\ H \text{ available}, \\ U^{s}_{i,\tau,a}, & 1950 \le \tau < 2015,\ U \text{ available}, \\ W^{s}_{i,\tau,a}, & \tau \ge 2015,\ W \text{ available}, \\ U^{s}_{i,\tau,a}, & \tau \ge 2015,\ W \text{ absent},\ U \text{ available}. \end{cases}$$

Years without an available target source are skipped. Under the supplied source coverage, this excludes 2021–2025 unless additional rasters are provided.

Let:

$$\mathcal{C}_{\tau} = \{i : P^{A}_{i,\tau} \ge 1\}.$$

When $|\mathcal{C}_\tau| > 15000$, deterministic stride subsampling selects:

$$i_j = \mathcal{C}_\tau \left[ \left\lfloor \frac{j|\mathcal{C}_\tau|}{15000} \right\rfloor \right], \qquad j = 0, \ldots, 14999.$$

Otherwise, all cells in $\mathcal{C}_\tau$ are candidates. Retain candidates with finite covariates and target total:

$$T_{i,\tau} = \sum_{s,a} T^s_{i,\tau,a} \geq 1.$$

Denote the resulting training set by $\mathcal{I}_\tau$, and define soft targets:

$$q^s_{i,\tau,a} = \frac{T^s_{i,\tau,a}}{T_{i,\tau}}.$$

## Model and identifiability

For anchor year $\tau$, the intercept-free multinomial model is:

$$p_a^{s,(\tau)}(X) = \frac{\exp(X^\top \beta^s_{\tau,a})}{\sum_{r,c} \exp(X^\top \beta^r_{\tau,c})}.$$

The reference cohort is `f_00`:

$$\boxed{\beta^f_{\tau,0} \equiv \mathbf{0}.}$$

Only the remaining 35 cohort parameter vectors are free.

## Weighted loss and feature scaling

Each training cell receives log-population weight:

$$w_{i,\tau} = \log(1 + T_{i,\tau}), \qquad W_\tau = \sum_{i \in \mathcal{I}_\tau} w_{i,\tau}.$$

For feature $j$, define its weighted RMS:

$$r_{\tau,j} = \sqrt{\frac{1}{W_\tau} \sum_{i \in \mathcal{I}_\tau} w_{i,\tau} \widetilde{X}^2_{i,\tau,j}}.$$

Replace $r_{\tau,j}$ by $1$ when it is at most $10^{-12}$. Set:

$$X'_{i,\tau,j} = \frac{\widetilde{X}_{i,\tau,j}}{r_{\tau,j}}, \qquad \beta^s_{\tau,a,j} = \frac{\widetilde{\beta}^s_{\tau,a,j}}{r_{\tau,j}}.$$

The fitted parameters approximately minimise:

$$\mathcal{L}_\tau(\widetilde{\beta}) = -\frac{1}{W_\tau}\sum_{i\in\mathcal{I}_\tau} w_{i,\tau}\sum_{s,a} q^s_{i,\tau,a}\log\max\left(p_a^{s,(\tau)}(\widetilde{X}_{i,\tau}),10^{-15}\right) + \frac{\lambda_{\mathrm{MNL}}}{2}\sum_{(s,a)\neq(f,0)}\|\widetilde{\beta}^s_{\tau,a}\|_2^2.$$

The supplied defaults are:

$$\lambda_{\mathrm{MNL}}=0, \qquad \eta_{\mathrm{learning}}=0.1, \qquad K_{\max}=50, \qquad \varepsilon_{\mathrm{fit}}=10^{-4}.$$

Batch gradient descent uses heavy-ball momentum:

$$v^{(k)} = 0.9\,v^{(k-1)} - \eta_{\mathrm{learning}}\nabla\mathcal{L}_\tau,$$

$$\widetilde{\beta}^{(k+1)} = \widetilde{\beta}^{(k)} + v^{(k)}.$$

The stopping test is:

$$\max_{c,j}|v^{(k)}_{c,j}| < 10^{-4},$$

in RMS-scaled parameter space, or termination at the iteration cap. This is a finite-iteration fit, not a guarantee that the exact minimiser has been attained.

# 1.3. Coverage-weighted temporal ensemble

Let $n_\tau$ be the saved training sample count, with fallback $1000$, and define:

$$n_{\max} = \max\left(1,\max_\tau n_\tau\right),$$

$$c_\tau = \left(\frac{n_\tau}{n_{\max}}\right)^{0.3}, \qquad \omega_\tau = \frac{c_\tau}{\sum_\nu c_\nu}.$$

The effective evaluation year is:

$$\boxed{t_{\mathrm{eff}} = \max(t,1750).}$$

Temporal kernel weights are:

$$v_\tau(t) = \frac{\omega_\tau e^{-|t_{\mathrm{eff}}-\tau|/50}}{\sum_\nu \omega_\nu e^{-|t_{\mathrm{eff}}-\nu|/50}}.$$

When a same-year model exists, use:

$$\alpha_\tau(t) = 0.8v_\tau(t) + 0.2\mathbf{1}_{\{\tau=t\}}.$$

Otherwise:

$$\alpha_\tau(t) = v_\tau(t).$$

A same-year model outside the loaded ensemble is assigned zero initial ensemble weight before receiving its $0.2$ contribution.

The ensemble prediction is represented as the probability mixture:

$$p^s_{i,t,a} = \sum_\tau \alpha_\tau(t) p_a^{s,(\tau)}(\widetilde{X}_{i,t}).$$

For valid, unmasked predictions:

$$p^s_{i,t,a} \geq 0, \qquad \sum_{s,a} p^s_{i,t,a} = 1.$$

For all $t \leq 1750$, the temporal kernel weights remain at their 1750 values while historical covariates continue to vary. There is no 1500 cutoff or separately loaded premodern model in this construction.[2]

## Uninhabited-density mask

Prediction uses `mask_uninhabited: true`. If `popd_[i] === 0`, then:

$$p^s_{i,t,a} = 0 \quad \text{for every } s, a.$$

Thus probability normalisation is not asserted on masked cells. If such a cell nevertheless has $P^A_{i,t} \geq 0.01$, downstream clamping uses the uniform fallback where required.

# 1.4. Country-bin aggregation

Country aggregation is active only when:

$$t > 1750 \quad \Longleftrightarrow \quad \gamma(t) > 0.$$

For each country bin $b$, define:

$$\mathcal{E}_{b,t} = \{i : b(i) = b,\ P^A_{i,t} \geq 0.01\}.$$

Pre-1950 HMD cells are not excluded: HMD replacement occurs later, during source compositing.

Preliminary country populations are:

$$R^s_{b,t,a} = \sum_{i \in \mathcal{E}_{b,t}} P^A_{i,t} p^s_{i,t,a}.$$

Define combined counts and annualised age densities:

$$R_{b,t,a} = R^{f}_{b,t,a} + R^{m}_{b,t,a},$$

$$d^{s}_{b,t,a} = \frac{R^{s}_{b,t,a}}{w_a}, \qquad d_{b,t,a} = \frac{R_{b,t,a}}{w_a}.$$

Below, suppress $b, t$ where convenient.

Because masked predictions may have zero total, $\sum_{s,a} R^{s}_{b,t,a}$ need not equal the full clamping population of the bin. This is one possible source of incompatibility between country targets and pixel margins in IPF.

# 1.5. PAVA graduation in the two spatial regimes

Let:

$$\mathcal{A}_{+} = \{2, \dots, 17\}.$$

Define male-share bounds from the biological male-to-female ratio bounds:

$$m^{-}_{a} = \frac{\rho^{-}_{a}}{1 + \rho^{-}_{a}}, \qquad m^{+}_{a} = \frac{\rho^{+}_{a}}{1 + \rho^{+}_{a}}.$$

## 1.5.1. Per-pixel regime: $t \leq 1750$

For $P^{A}_{i,t} < 0.01$, write all clamped cohorts as zero. Otherwise, perform the following operations.

**Step 1 — Elderly monotonic ceiling.**

Initialise:

$$\widetilde{p}^{s}_{i,a} = [p^{s}_{i,t,a}]_{+}.$$

Sequentially for $a = 14, 15, 16, 17$, corresponding to $65, 70, 75, 80+$, if:

$$\widetilde{p}^{s}_{i,a-1} > 0,$$

replace:

$$\widetilde{p}^{s}_{i,a} \leftarrow \min\left(\widetilde{p}^{s}_{i,a}, 1.05\,\frac{w_a}{w_{a-1}}\widetilde{p}^{s}_{i,a-1}\right).$$

**Step 2 — Coupled combined-density PAVA and sex allocation.**

Form:

$$d_{i,a} = \frac{\widetilde{p}^{f}_{i,a} + \widetilde{p}^{m}_{i,a}}{w_a}.$$

Apply weighted decreasing PAVA on $\mathcal{A}_+$:

$$\widehat{d}_i = \operatorname*{argmin}_{z_2 \geq \cdots \geq z_{17} \geq 0} \sum_{a=2}^{17} w_a (z_a - d_{i,a})^2.$$

Keep the combined counts for $a < 2$, and use $w_a \widehat{d}_{i,a}$ for $a \geq 2$. Allocate each combined cohort between sexes using its empirical male share clipped to $[m_a^-, m_a^+]$. A zero combined cohort carries no mass, so its allocation share does not affect the result.

**Step 3 — Secondary sex-specific PAVA.**

Apply decreasing PAVA separately to the female and male cohort buffers on $\mathcal{A}_+$, with weights $w_a$.

Since $w_a = 5$ throughout this suffix, decreasing cohort counts and decreasing annualised densities are equivalent there. This operation guarantees decreasing per-sex suffix density, but can perturb the sex-ratio bounds imposed in Step 2.

**Step 4 — Per-pixel normalisation.**

Let $C^s_{i,a}$ denote the resulting cohort buffers. Set:

$$N^{s,\mathrm{clamp}}_{i,t,a} = \begin{cases} P^A_{i,t} \dfrac{C^s_{i,a}}{\sum_{r,c} C^r_{i,c}}, & \sum_{r,c} C^r_{i,c} > 0, \\ P^A_{i,t}/36, & \text{otherwise.} \end{cases}$$

Hence:

$$\sum_{s,a} N^{s,\mathrm{clamp}}_{i,t,a} = P^A_{i,t}.$$

The original female and male totals are not separately preserved.

## 1.5.2. Country-bin PAVA endpoint: $t > 1750$

For country-bin combined densities, define:

$$\widehat{d}^I = \operatorname*{argmin}_{z_2 \geq \cdots \geq z_{17} \geq 0} \sum_{a=2}^{17} w_a (z_a - d_a)^2.$$

The combined endpoint is:

$$G_a^I = \begin{cases} R_a, & a < 2, \\ w_a \widehat{d}_a^I, & a \geq 2. \end{cases}$$

For $R_a > 0$, let $m_a = R_a^m / R_a$, and set:

$$m_a^I = \mathrm{clip}(m_a; m_a^-, m_a^+),$$

$$G_a^{m,I} = G_a^I m_a^I, \qquad G_a^{f,I} = G_a^I (1 - m_a^I).$$

A fallback share is used where the empirical share is undefined.

For a pooled PAVA block $\mathcal{B}$:

$$\bar{d}_{\mathcal{B}} = \frac{\sum_{a \in \mathcal{B}} w_a d_a}{\sum_{a \in \mathcal{B}} w_a},$$

so:

$$\sum_{a \in \mathcal{B}} w_a \bar{d}_{\mathcal{B}} = \sum_{a \in \mathcal{B}} R_a.$$

Consequently:

$$\boxed{\sum_{s,a} G_a^{s,I} = \sum_{s,a} R_a^s.}$$

This is **combined-sex conservation**, not separate conservation by sex. Because male shares vary with age, individual-sex densities need not be decreasing.

Unlike the per-pixel regime, this country-bin endpoint has no secondary sex-specific PAVA.

# 1.6. Coupled Huber–Whittaker endpoint

The Whittaker endpoint graduates:

1. combined age density on $\mathcal{A}_+$;
2. male shares across all ages.

Define Huber loss:

$$\rho_\delta(e) = \begin{cases} \frac{1}{2} e^2, & |e| \leq \delta, \\ \delta |e| - \frac{1}{2} \delta^2, & |e| > \delta. \end{cases}$$

For positive combined suffix mass, let:

$$\bar{d} = \frac{\sum_{a=2}^{17} R_a}{\sum_{a=2}^{17} w_a}, \qquad y_a = \frac{d_a}{\bar{d}}.$$

The robust density solver is represented by:

$$\hat{z} \approx \operatorname*{argmin}_{z_a \geq 0} \left\{ \sum_{a=2}^{17} w_a \rho_\delta(z_a - y_a) + \frac{\lambda_d}{2} \sum_{a=2}^{15} (\Delta^2 z_a)^2 + \frac{\mu_d}{2} \sum_{a=2}^{16} [\Delta z_a]_+^2 \right\},$$

where:

$$\Delta z_a = z_{a+1} - z_a, \qquad \Delta^2 z_a = z_{a+2} - 2z_{a+1} + z_a.$$

The active orchestrator defaults are:

$$\delta = 0.05, \qquad \lambda_d = 20, \qquad \mu_d = 5000.$$

The monotonicity penalty is soft: it penalises increasing density but does not impose an exact isotonic constraint. The coupling helper's direct-call default for $\mu_d$ is $0$, distinct from the orchestrated setting.

Coordinate descent uses at most 200 outer sweeps, with 30-step inner coordinate bisection. The stated convergence test uses maximum absolute KKT gradient residual below $10^{-5}$.[3]

Restore combined suffix population:

$$c_d = \frac{\sum_{a=2}^{17} R_a}{\sum_{a=2}^{17} w_a \bar{d}\, \hat{z}_a},$$

$$G_a^W = \begin{cases} R_a, & a < 2, \\ c_d w_a \bar{d}\, \hat{z}_a, & a \geq 2. \end{cases}$$

This normalisation requires positive finite provisional suffix mass. Empty suffixes remain zero.

## Male-share graduation

Use empirical shares when $R_a > 10^{-9}$; otherwise use:

$$m_a = \frac{\rho_a^{\text{base}}}{1 + \rho_a^{\text{base}}},$$

with baseline ratios:

$$\begin{aligned} \rho^{\text{base}} = (&1.045, 1.042, 1.040, 1.038, 1.030, 1.015, \\ &0.985, 0.975, 0.965, 0.960, 0.950, 0.930, \\ &0.900, 0.860, 0.810, 0.750, 0.680, 0.600). \end{aligned}$$

Unless fixed empirical ratios are requested, the share solver is represented by:

$$\widehat{m} \approx \operatorname*{argmin}_{u_a \geq 0} \left\{ \sum_{a=0}^{17} w_a \rho_{0.05}(u_a - m_a) + \frac{20}{2} \sum_{a=0}^{15} (\Delta^2 u_a)^2 \right\}.$$

Biological ratio bounds are enforced in the active default configuration:

$$m_a^W = \mathrm{clip}(\widehat{m}_a; m_a^-, m_a^+).$$

The supplied ratio bounds are:

$$[\rho_a^-, \rho_a^+] = \begin{cases} [1.02, 1.07], & a = 0, \\ [1.01, 1.07], & a = 1, \\ [0.95, 1.10], & a = 2, 3, \\ [0.50, 1.80], & a = 4, \dots, 13, \\ [0.30, 1.05], & a = 14, \dots, 17. \end{cases}$$

Allocate:

$$G_a^{m,W} = G_a^W m_a^W, \qquad G_a^{f,W} = G_a^W (1 - m_a^W).$$

Then:

$$\boxed{\sum_{s,a} G_a^{s,W} = \sum_{s,a} R_a^s.}$$

Separate female and male totals need not be preserved. If bounds are explicitly disabled, validity additionally requires the resulting shares to lie in $[0, 1]$.

# 1.7. Logistic demographic-transition kernel

Let:

$$\sigma(u) = \frac{1}{1 + e^{-u}}.$$

Define:

$$\gamma(t) = \begin{cases} 0, & t \leq 1750, \\ \dfrac{\sigma\left(8\left(\dfrac{t - 1750}{100} - \frac{1}{2}\right)\right) - \sigma(-4)}{\sigma(4) - \sigma(-4)}, & 1750 < t < 1850, \\ 1, & t \geq 1850. \end{cases}$$

Thus:

$$0 \le \gamma(t) \le 1, \qquad \gamma(1800) = \frac{1}{2}.$$

For $t \le 1750$, execution follows the per-pixel operator of §1.5.1, not country-bin PAVA.

For $t > 1750$, define the formal endpoint blend:

$$G^{s,\mathrm{blend}}_{b,t,a} = [1-\gamma(t)]G^{s,I}_{b,t,a} + \gamma(t)G^{s,W}_{b,t,a}.$$

When its combined cohort mass is positive, the blended male share is:

$$\frac{(1-\gamma)G^I_a m^I_a + \gamma G^W_a m^W_a}{(1-\gamma)G^I_a + \gamma G^W_a}.$$

This is a convex combination of the endpoint shares, with mass-dependent weights. Therefore it lies in $[m^-_a, m^+_a]$. Neither endpoint, however, guarantees decreasing density separately by sex.

## Implemented scale-factor form and support restriction

The implementation forms:

$$k^{s,\bullet}_{b,t,a} = \begin{cases} G^{s,\bullet}_{b,t,a}/R^s_{b,t,a}, & R^s_{b,t,a} > 0, \\ 1, & R^s_{b,t,a} = 0, \end{cases} \qquad \bullet \in \{I, W\},$$

$$k^{s,(0)}_{b,t,a} = [1-\gamma(t)]k^{s,I}_{b,t,a} + \gamma(t)k^{s,W}_{b,t,a},$$

and the implemented redistribution target is:

$$\boxed{G^s_{b,t,a} = R^s_{b,t,a}k^{s,(0)}_{b,t,a}.}$$

For positive $R^s_{b,t,a}$, this equals the formal blend. If $R^s_{b,t,a} = 0$, the implemented target remains zero even if endpoint graduation would assign positive mass.

Consequently, formal endpoint mass-conservation and sex-ratio results transfer directly to the implemented target only where this support restriction does not remove positive endpoint mass. Subsequent pixel normalisation, finite IPF, and source compositing introduce further distinctions between endpoint properties and final raster properties.

# 1.8. Three-pass IPF redistribution and source compositing

## Country-to-cell redistribution

For $t > 1750$, perform passes $\ell = 0, 1, 2$. On $i \in \mathcal{E}_{b,t}$, define:

$$Z_{i,t}^{(\ell)} = \sum_{r,c} p_{i,t,c}^{r} k_{b(i),t,c}^{r,(\ell)}.$$

When $Z_{i,t}^{(\ell)} > 0$:

$$N_{i,t,a}^{s,(\ell)} = P_{i,t}^{A} \frac{p_{i,t,a}^{s} k_{b(i),t,a}^{s,(\ell)}}{Z_{i,t}^{(\ell)}}.$$

When the denominator is zero, use:

$$N_{i,t,a}^{s,(\ell)} = \begin{cases} 0, & \ell = 0, 1, \\ P_{i,t}^{A}/36, & \ell = 2. \end{cases}$$

For $\ell = 0, 1$, calculate:

$$S_{b,t,a}^{s,(\ell)} = \sum_{i \in \mathcal{E}_{b,t}} N_{i,t,a}^{s,(\ell)},$$

and, when $S_{b,t,a}^{s,(\ell)} > 0$, update:

$$k_{b,t,a}^{s,(\ell+1)} = k_{b,t,a}^{s,(\ell)} \frac{G_{b,t,a}^{s}}{S_{b,t,a}^{s,(\ell)}}.$$

Otherwise retain the factor. The final clamped output is:

$$N_{i,t,a}^{s,\mathrm{clamp}} = \begin{cases} N_{i,t,a}^{s,(2)}, & P_{i,t}^{A} \geq 0.01, \\ 0, & P_{i,t}^{A} < 0.01. \end{cases}$$

Three-pass IPF actively rakes country cohort sums towards the targets while restoring pixel population margins on each nonzero-denominator pass. It does not establish exact country margins after only three passes.

Multiplicative raking cannot generally create absent support. The final uniform fallback can introduce mass outside the multiplicative support and can further disturb country targets.

## Step F: source compositing

For $t < 1950$, where HMD rasters exist and:

$$H_{i,t}^{\Sigma} = \sum_{r,c} H_{i,t,c}^{r} > 0,$$

the final compositor uses:

$$N^{s}_{i,t,a} = \begin{cases} P^{A}_{i,t}\dfrac{H^{s}_{i,t,a}}{H^{\Sigma}_{i,t}}, & P^{A}_{i,t} > 0, \\[2ex] H^{s}_{i,t,a}, & P^{A}_{i,t} \le 0. \end{cases}$$

Other pre-1950 cells retain their clamped modelled output.

For $t \ge 1950$, selection by available source raster is:

$$N^{s}_{i,t,a} = \begin{cases} U^{s}_{i,t,a}, & 1950 \le t < 2015,\ U \text{ available}, \\ W^{s}_{i,t,a}, & t \ge 2015,\ W \text{ available}, \\ U^{s}_{i,t,a}, & t \ge 2015,\ W \text{ absent},\ U \text{ available}, \\ N^{s,\mathrm{clamp}}_{i,t,a}, & \text{otherwise}. \end{cases}$$

Under the supplied default coverage, the last branch governs 2021–2025. Source replacement occurs after graduation and can replace its shape constraints.

# 1.9. Stock-reconciliation theorem and exceptions

For every finite, successfully clamped cell with $P^{A}_{i,t} \ge 0.01$:

$$\boxed{\sum_{s,a} N^{s,\mathrm{clamp}}_{i,t,a} = P^{A}_{i,t}.}$$

The same identity holds for positive-population HMD-normalised cells.

UNWPP cohorts are dasymetrically clamped to the Stadestér/Velkscala stock. On mapped populated pixels in 1950–2020:

$$\sum_{s,a} U^{s}_{i,t,a} = P_{i,t}.$$

The UNWPP preprocessing also replaces national infant male-to-female ratios outside $[0.90, 1.20]$ by $1.05$ before the relevant scaling.

WorldPop cohorts in 2015–2020 are copied without this population clamp. Therefore:

$$\sum_{s,a} W^{s}_{i,t,a} = P_{i,t}$$

is **not guaranteed**; inequality is possible, not logically necessary in every cell.

For the final composite, define the stock residual:

$$e^{N}_{i,t} = \sum_{s,a} N^{s}_{i,t,a} - P_{i,t}.$$

The principal structural exceptions to $e^N_{i,t} = 0$ are:

1. **Unclamped WorldPop selections**, normally 2015–2020.
2. **Modelled outputs in 2024–2025**, clamped to $P_{i,2023}$ rather than $P_{i,t}$.
3. **Modelled cells with** $0 < P^A_{i,t} < 0.01$, which are zeroed unless subsequently replaced by a source.
4. **Pre-1950 HMD cells with** $P^A_{i,t} = 0$ **but positive HMD mass**, where the source counts pass through unscaled.

Subject to source coverage, support, finite inputs, and these exceptions:

$$e^N_{i,t} = 0 \quad \Longrightarrow \quad \sum_i \sum_{s,a} N^s_{i,t,a} = \sum_i P_{i,t}.$$

Floating-point storage adds numerical residuals to these structural qualifications.

---

# Definition 2. Migration estimation, calibration, and global conservation

## 2.1. Notation

For year $t$, let:

- $A^B$, $A^{D_f}$, $A^{D_m}$ denote source vital counts;
- $g_{i,t}$ denote the annualised slope stored in the population-delta raster;
- $\widetilde{M}_{i,t}$ denote preliminary migration before waterfill;
- $M^0_{i,t}$ denote migration passed to final reconciliation;
- $M^{s,0}_{i,t}$ denote its sex allocation;
- $\Omega_+(t) = \{i : P_{i,t} > 0\}$.

For $v \in \{M, B, D_f, D_m\}$, let:

- $Y^v$ denote regression targets;
- $z^v$ denote raw OLS predictions;
- $u^v \in [0, 1]$ denote normalised predictions;
- $f^v$ denote calibrated rates or factors;
- $\theta^v_\tau$ denote year-specific coefficients.

The design vector is shared with age–sex estimation:

$$Z_{i,\tilde{t}} \equiv \widetilde{X}_{i,t} \in \mathbb{R}^{35}.$$

There is no population-delta raster at the initial year $-10000$. Branches that substitute $g = 0$ when a delta raster is absent do not thereby establish an observed interval change.

## 2.2. Unclipped migration targets

For target construction, prefer valid Niva/Kummu vital counts where available, normally 2000–2019; otherwise use UNWPP:

$$(A^B_{i,t}, A^D_{i,t}) = \begin{cases} (A^{B,K}_{i,t}, A^{D,K}_{i,t}), & \text{valid Niva/Kummu pair available,} \\ (A^{B,U}_{i,t}, A^{D_f,U}_{i,t} + A^{D_m,U}_{i,t}), & \text{otherwise, with valid UNWPP births.} \end{cases}$$

Negative vital counts are sanitised to zero. The UNWPP fallback also replaces missing sex-specific deaths by zero. Missing population-delta pixels default to $g_{i,t} = 0$ in target construction.

Define:

$$A^M_{i,t} = g_{i,t} - A^B_{i,t} + A^D_{i,t}.$$

For available target years $t \geq 1950$ and $P_{i,t} \geq 1$:

$$\boxed{Y^M_{i,t} = \frac{g_{i,t} - A^B_{i,t} + A^D_{i,t}}{P_{i,t}}.}$$

There is no clipping to $[-1, 1]$. These are residual migration targets, not independent observations of migration.

## 2.3. Intercept-free RMS-scaled ridge regression

Retain cells with finite targets, finite covariates, and:

$$Y^v_{i,\tau} \neq 0 \quad \text{or} \quad Z_{i,\tilde{\tau}} \neq \mathbf{0}.$$

Let their set be $\mathcal{I}^v_\tau$. Define feature RMS values:

$$s_j = \sqrt{\frac{1}{|\mathcal{I}^v_\tau|} \sum_{i \in \mathcal{I}^v_\tau} Z^2_{i,\tilde{\tau},j}},$$

with fallback $s_j = 1$ when the calculated RMS is at most $10^{-12}$.

There is no intercept. Scaling columns by $s_j$ and applying ridge penalty $\lambda_v I$ is equivalent, in original coordinates, to:

$$\hat{\theta}^v_\tau = \underset{\theta\in\mathbb{R}^{35}}{\operatorname{argmin}} \left\{ \sum_{i\in\mathcal{I}^v_\tau} \left( Y^v_{i,\tau} - Z^{\mathsf{T}}_{i,\widetilde{\tau}}\theta \right)^2 + \lambda_v \theta^{\mathsf{T}} J^v_\tau \theta \right\},$$

where:

$$\boxed{J^v_\tau = \operatorname{diag}(s_1^2, \ldots, s_{35}^2), \qquad \lambda_v = 1.}$$

Prediction uses:

$$\theta^v(t) = \begin{cases} \hat{\theta}^v_t, & \text{same-year model exists,} \\ \overline{\theta}^v, & \text{otherwise,} \end{cases}$$

$$z^v_{i,t} = Z^{\mathsf{T}}_{i,\tilde{t}} \theta^v(t).$$

## Fallback coefficient aggregation

For coefficient $j$, let $\mathcal{P}_j$ and $\mathcal{N}_j$ index finite positive and negative anchor coefficients. Let $K_{\text{clean},j}$ count all finite coefficients, including exact zeros.

For $K_{\text{clean},j} > 0$:

$$\boxed{\overline{\theta}^v_j = \left| \frac{|\mathcal{P}_j|}{K_{\text{clean},j}} \exp\left( \frac{1}{|\mathcal{P}_j|} \sum_{\tau\in\mathcal{P}_j} \log \hat{\theta}^v_{\tau,j} \right) - \frac{|\mathcal{N}_j|}{K_{\text{clean},j}} \exp\left( \frac{1}{|\mathcal{N}_j|} \sum_{\tau\in\mathcal{N}_j} \log |\hat{\theta}^v_{\tau,j}| \right) \right|.}$$

Empty sign groups contribute zero. Exact zeros contribute to the denominator but neither geometric mean.

The supplied `Math.abs` function is applied **after** signed geometric aggregation. Hence:

$$\overline{\theta}^v_j \geq 0.$$

This does not guarantee non-negative raw predictions when covariates can be negative. The formula also presumes at least one finite contributing coefficient; it does not specify an otherwise undocumented empty-model fallback.

# 2.4. Target-bound calibration and spatial normalisation

For a nonempty sorted sample $y_{(0)}, \ldots, y_{(n-1)}$, use:

$$Q_p(y) = y_{(\lfloor np \rfloor)}, \qquad 0 < p < 1.$$

Migration and vital events have distinct target-bound rules.

## 2.4a. Migration bounds: annual log-tail Tukey compression

Construct a stride-17 sample using raster indices $0, 17, 34, \ldots$:

- If a migration-target raster exists, sample finite target values on positive-population cells; valid target construction requires $P \geq 1$.
- Otherwise, if a population-delta raster exists, sample finite $g_{i,t}/P_{i,t}$ on cells with $P_{i,t} > 0$.
- If neither supplies observations, the sample is empty.

For nonempty sample $S_t$, let:

$$q_1^s = Q_{0.25}(S_t), \qquad q_3^s = Q_{0.75}(S_t), \qquad I_s = q_3^s - q_1^s,$$

$$F_s^- = q_1^s - 1.5 I_s, \qquad F_s^+ = q_3^s + 1.5 I_s.$$

Define:

$$R_s(x) = \begin{cases} F_s^+ + I_s \log\left(1 + \dfrac{x - F_s^+}{I_s}\right), & I_s > 0,\ x > F_s^+, \\ F_s^- - I_s \log\left(1 + \dfrac{F_s^- - x}{I_s}\right), & I_s > 0,\ x < F_s^-, \\ x, & \text{otherwise.} \end{cases}$$

Set:

$$L_t^M = R_s(\min S_t), \qquad U_t^M = R_s(\max S_t).$$

If $U_t^M \leq L_t^M$, use $(F_s^-, F_s^+)$. For an empty sample:

$$L_t^M = U_t^M = 0.$$

Migration calibration does not use the former population-above-10000 filter, near-zero-target exclusion, 1st/99th percentiles, cross-year bound pooling, or exponential local/global shrinkage.

The growth-rate fallback applies whenever no migration-target raster exists, including historical years and potentially years after empirical target coverage.

## 2.4b. Vital-event bounds: local/global quantile shrinkage

For $v \in \{B, D_f, D_m\}$, retain positive targets at stride 7 and calculate:

$$\ell_t^v = Q_{0.01}(Y_t^v), \qquad u_t^{v,\text{bound}} = Q_{0.99}(Y_t^v).$$

Across available target statistics:

$$\ell_*^v = \min_t \ell_t^v, \qquad u_*^{v,\text{bound}} = \max_t u_t^{v,\text{bound}}.$$

Let:

$$n_t^v = 7 \times (\text{retained sample count}), \qquad \varepsilon_t^v = e^{-n_t^v/15}.$$

Then:

$$L_t^v = \max\left(0,\ (1-\varepsilon_t^v)\ell_t^v + \varepsilon_t^v \ell_*^v\right),$$

$$U_t^v = (1-\varepsilon_t^v)u_t^{v,\text{bound}} + \varepsilon_t^v u_*^{v,\text{bound}}.$$

Without same-year statistics, use the global bounds. This construction assumes that usable global target statistics exist.

## 2.4c. Normalisation of raw OLS predictions

On valid inhabited land:

$$\text{landarea}_i > 0, \qquad P_{i,t} > 0,$$

calculate:

$$q_1 = Q_{0.25}(z_t^v), \qquad q_3 = Q_{0.75}(z_t^v), \qquad I = q_3 - q_1.$$

Let $\sigma_z$ be the population standard deviation. Set:

$$a = \begin{cases} I, & I > 10^{-5}, \\ \sigma_z, & I \le 10^{-5},\ \sigma_z > 10^{-5}, \\ 1, & \text{otherwise and } v = M, \\ 0.01, & \text{otherwise and } v \in \{B, D_f, D_m\}. \end{cases}$$

With:

$$T^- = q_1 - 1.5a, \qquad T^+ = q_3 + 1.5a,$$

define:

$$R(z) = \begin{cases} T^+ + a\log\left(1 + \dfrac{z - T^+}{a}\right), & z > T^+, \\ T^- - a\log\left(1 + \dfrac{T^- - z}{a}\right), & z < T^-, \\ z, & \text{otherwise.} \end{cases}$$

Writing $R_{\min}$ and $R_{\max}$ for the extrema over the valid domain:

$$u_{i,t}^{v} = \begin{cases} \operatorname{clip}\left( \dfrac{R(z_{i,t}^{v}) - R_{\min}}{R_{\max} - R_{\min}}; 0, 1 \right), & R_{\max} > R_{\min}, \\ 1/2, & R_{\max} = R_{\min}. \end{cases}$$

Finally:

$$\boxed{f_{i,t}^{v} = L_{t}^{v} + u_{i,t}^{v}(U_{t}^{v} - L_{t}^{v}).}$$

Thus, for ordered finite bounds:

$$L_{t}^{v} \leq f_{i,t}^{v} \leq U_{t}^{v}.$$

Denote the complete variable-specific calibration by $\mathcal{C}_{v,t}$.[4]

# 2.5. Preliminary migration, feasibility ceiling, and waterfill

Outside $\Omega_{+}(t)$, migration is zero.

## Preliminary migration

If all three UNWPP vital rasters and the delta raster exist, define capped total deaths:

$$A_{i,t}^{D,U} = \min\left( P_{i,t}, [A_{i,t}^{D_f,U}]_{+} + [A_{i,t}^{D_m,U}]_{+} \right).$$

Then:

$$\widetilde{M}_{i,t} = \min\left( g_{i,t} - [A_{i,t}^{B,U}]_{+} + A_{i,t}^{D,U},\ g_{i,t} + P_{i,t} \right).$$

With non-negative births and the death cap, this ceiling is algebraically redundant in exact arithmetic for the empirical residual, but remains an explicit implementation safeguard.

Otherwise:

$$\widetilde{M}_{i,t} = \min\left( P_{i,t} f_{i,t}^{M},\ g_{i,t} + P_{i,t} \right),$$

using $g_{i,t} = 0$ if no delta raster exists.

The final empirical-generation branch uses UNWPP even where regression targets preferred Niva/Kummu data.

## Three-iteration one-sided waterfill

Initialise:

$$M_{i,t}^{(0)} = \widetilde{M}_{i,t}.$$

For $k = 0, 1, 2$, define:

$$S_t^{(k)} = \sum_{i \in \Omega_+(t)} M_{i,t}^{(k)},$$

$$\mathcal{U}_t^{(k)} = \{ i \in \Omega_+(t) : M_{i,t}^{(k)} < g_{i,t} + P_{i,t} \},$$

$$P_{\mathcal{U}^{(k)},t} = \sum_{i \in \mathcal{U}_t^{(k)}} P_{i,t}.$$

Terminate early if:

$$|S_t^{(k)}| < 0.01 \quad \text{or} \quad P_{\mathcal{U}^{(k)},t} = 0.$$

Otherwise set:

$$\kappa_t^{(k)} = -\frac{S_t^{(k)}}{P_{\mathcal{U}^{(k)},t}},$$

and update:

$$M_{i,t}^{(k+1)} = \begin{cases} \min\left( M_{i,t}^{(k)} + P_{i,t}\kappa_t^{(k)}, g_{i,t} + P_{i,t} \right), & i \in \mathcal{U}_t^{(k)}, \\ M_{i,t}^{(k)}, & i \notin \mathcal{U}_t^{(k)}. \end{cases}$$

Denote the final iterate by $M_{i,t}^{0}$.

## Accounting properties

The ceiling is preserved:

$$\boxed{M_{i,t}^{0} \le g_{i,t} + P_{i,t}.}$$

Define the remaining global residual:

$$\varepsilon_{0,t}^{M} = \sum_i M_{i,t}^{0}.$$

The algorithm targets:

$$|\varepsilon_{0,t}^{M}| < 0.01,$$

but the tolerance is not an unconditional theorem. It can fail because the eligible set is empty or because additional saturation rounds would be required after the three permitted updates.

Cells already at their ceiling remain excluded even when $S_t^{(k)} > 0$, although lowering those cells would be feasible. Thus this is not a general exact projection onto the globally balanced feasible set.

## 2.6. Sex allocation

For the empirical branch:

$$M_{i,t}^{f,0} = 0.488 M_{i,t}^0, \qquad M_{i,t}^{m,0} = 0.512 M_{i,t}^0.$$

These are fixed SRB-based allocation shares, not estimates of the actual sex composition of migration.

For the modelled branch, use the reduced Rogers–Castro schedule:

$$W_s(x) = 0.02 e^{-0.1x} + 0.06 \exp\left[-0.1(x-\mu_s) - e^{-0.3(x-\mu_s)}\right] + 0.003,$$

with:

$$\mu_f = 20, \qquad \mu_m = 22.5.$$

Define:

$$V_{i,t}^s = \sum_a [N_{i,t,a}^s]_+ W_s(x_a),$$

$$\pi_{f,i,t}^M = \begin{cases} \dfrac{V_{i,t}^f}{V_{i,t}^f + V_{i,t}^m}, & V_{i,t}^f + V_{i,t}^m > 0, \\ 0.488, & \text{otherwise}, \end{cases}$$

$$\pi_{m,i,t}^M = 1 - \pi_{f,i,t}^M, \qquad M_{i,t}^{s,0} = \pi_{s,i,t}^M M_{i,t}^0.$$

Therefore:

$$\boxed{M_{i,t}^{f,0} + M_{i,t}^{m,0} = M_{i,t}^0.}$$

These schedules allocate signed net migration. They do not separately identify immigration and emigration, and do not impose separate global female and male balances.

## 2.7. Relation to local demographic accounting

The local accounting identity is:

$$g_{i,t} = B_{i,t} - D_{i,t} + M_{i,t}.$$

For the capped empirical baseline:

$$\begin{aligned}&[A_{i,t}^{B,U}]_+ - A_{i,t}^{D,U} + M_{i,t}^0 - g_{i,t} \\ &\quad = M_{i,t}^0 - \left(g_{i,t} - [A_{i,t}^{B,U}]_+ + A_{i,t}^{D,U}\right).\end{aligned}$$

The right-hand side includes the effects of any preliminary ceiling adjustment and subsequent waterfill. It is not generally a single common population-weighted shift.

Definition 3 closes this local residual by adjusting unlocked vital counts and, where required, migration.

---

# Definition 3. Birth/death estimation and bivariate hyperbolic reconciliation

## 3.1. Notation

For a fixed year, suppress $t$ where convenient. Let:

- $B_i^0, D_{f,i}^0, D_{m,i}^0$ denote preliminary vital counts;
- $D_i^0 = D_{f,i}^0 + D_{m,i}^0$;
- $M_i^0$ denote migration from Definition 2;
- $\ell_{B,i}, \ell_{f,i}, \ell_{m,i} \in \{0,1\}$ denote source-override indicators;
- $h_i \in \{0,1\}$ denote the effective HMD whole-cell lock;
- $\mathrm{emp}_t$ indicate availability of all three UNWPP vital rasters;
- $b(i)$ denote the ISO3 bin, with unmapped cells assigned to `GLOBAL`;
- $s_b^B, s_b^D$ denote country scaling factors;
- $\Delta_b, \Delta_i$ denote hyperbolic discriminants;
- $B_i^*, D_i^*, M_i^*$ denote final reconciled totals;
- $Q_i^B, Q_i^{D_s}$ denote vital-event predictors.

The nominal reconciled domain is:

$$\mathcal{R}_t = \{i : P_i > 0,\ g_i \text{ finite},\ M_i^0 \text{ finite}\}.$$

Finite, non-negative preliminary vital counts are also required for the mathematical constructions below.

## 3.2. Birth predictor

Define:

$$\boxed{Q^B_{i,t} = N^f_{i,t,0} + N^m_{i,t,0}.}$$

Under age–sex stock reconciliation:

$$0 \le Q^B_{i,t} \le P_{i,t}.$$

Where age–sex stock reconciliation fails, the unconditional bound is instead:

$$0 \le Q^B_{i,t} \le \sum_{s,a} N^s_{i,t,a}.$$

## 3.3. Death predictors

On positive-current-population cells, define recent local growth:

$$r_{i,t} = \begin{cases} \dfrac{\log(P_{i,t}/P_{i,t^-})}{h_t}, & P_{i,t^-} > 0, \\ 0, & \text{otherwise.} \end{cases}$$

If the preceding population raster is unavailable, substitute current population, giving $r_{i,t} = 0$.

Construct growth-adjusted age density:

$$S^s_{i,t,a} = \frac{N^s_{i,t,a}}{w_a} e^{r_{i,t} x_a}.$$

This is a survivorship-shaped demographic proxy, not independently observed survival.

Apply weighted decreasing isotonic regression over all ages:

$$\widehat{S}^s_{i,t} = \operatorname*{argmin}_{z_0 \ge \cdots \ge z_{17} \ge 0} \sum_{a=0}^{17} w_a (z_a - S^s_{i,t,a})^2.$$

This mortality-proxy PAVA is distinct from the age–sex graduation operator and applies in every modelled year.

For $a < 17$:

$$\widehat{\mu}^s_{i,t,a} = \begin{cases} \dfrac{\widehat{S}^s_{i,t,a} - \widehat{S}^s_{i,t,a+1}}{\widehat{S}^s_{i,t,a}(x_{a+1} - x_a)}, & \widehat{S}^s_{i,t,a} > 0, \\ 0, & \text{otherwise.} \end{cases}$$

For the terminal band:

$$\widehat{\mu}^{s}_{i,t,17} = \begin{cases} 1/5, & \widehat{S}^{s}_{i,t,17} > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Then:

$$\boxed{Q^{D_s}_{i,t} = \sum_{a} N^{s}_{i,t,a} \widehat{\mu}^{s}_{i,t,a}.}$$

Monotonicity gives:

$$0 \le \widehat{\mu}^{s}_{i,t,a} \le \frac{1}{x_{a+1} - x_a} \quad (a < 17).$$

Since the midpoint gaps are $2.5, 4.5, 5, \dots, 5$:

$$0 \le Q^{D_s}_{i,t} \le 0.4 \sum_{a} N^{s}_{i,t,a}.$$

These bounds presume finite inputs and finite exponential transformations. The proxy does not independently identify mortality or exactly remove migration effects.

## 3.4. Vital-event targets and preliminary counts

Use HMD sources before 1950 and UNWPP thereafter. On valid target cells:

$$Y^{B}_{i,t} = \min\left(1.5, \frac{A^{B}_{i,t}}{Q^{B}_{i,t}}\right),$$

$$Y^{D_s}_{i,t} = \min\left(3, \frac{A^{D_s}_{i,t}}{Q^{D_s}_{i,t}}\right).$$

Target filters require:

$$A^{v}_{i,t} > 0, \qquad Q^{v}_{i,t} \ge 1,$$

and, when the population raster exists:

$$P_{i,t} \ge 10000.$$

Fit and calibrate using the ridge construction in §2.3 and the vital-event bounds in §2.4b:

$$f^{v}_{i,t} = \mathcal{C}_{v,t}\left(Z^{\top}_{i,\tilde{t}} \theta^{v}(t)\right).$$

On the ordinary modelled/source-override branch, positive source values replace modelled counts. Without such an override:

$$B_i^0 = \min(f_i^B Q_i^B, 1.5 Q_i^B),$$

$$\widetilde{D}_i^s = \min(f_i^{D_s} Q_i^{D_s}, P_i).$$

If neither sex-specific death component has a positive source override, define:

$$\pi_{m,i}^D = \begin{cases} \dfrac{u_i^{D_m}}{u_i^{D_m} + u_i^{D_f}}, & u_i^{D_m} + u_i^{D_f} > 0, \\ 1/2, & \text{otherwise.} \end{cases}$$

Then:

$$D_{m,i}^0 = \pi_{m,i}^D (\widetilde{D}_i^m + \widetilde{D}_i^f),$$

$$D_{f,i}^0 = (1 - \pi_{m,i}^D)(\widetilde{D}_i^m + \widetilde{D}_i^f).$$

If either death component is overridden, this redistribution is skipped.

When all three UNWPP vital files exist, the empirical branch instead initialises:

$$(B_i^0, D_{f,i}^0, D_{m,i}^0) = (A_i^{B,U}, A_i^{D_f,U}, A_i^{D_m,U}),$$

after non-negative sanitation, and skips the preliminary sex redistribution.

The caps in this subsection constrain preliminary modelled counts. They are not all retained as final reconciliation constraints.

## 3.5. Effective locks and admissible flow domain

Let $\mathrm{emp}_t = 1$ if all three UNWPP vital rasters exist; otherwise let it be zero. Define:

$$\boxed{h_i = \mathbf{1}\left\{\mathrm{emp}_t = 0 \ \wedge \ [\ell_{B,i} = 1 \ \vee \ (\ell_{f,i} = 1 \wedge \ell_{m,i} = 1)]\right\}.}$$

Under the supplied source-selection rules, these are the pre-1950 HMD locks.

The effective lock is a **whole-cell vital lock**, not an independent lock for each flow:

- If births are HMD-overridden, all preliminary vital counts on that cell are retained, including modelled deaths.
- If both sex-specific deaths are HMD-overridden, all preliminary vital counts are retained, including modelled births.
- A source override for only one death sex, without a birth override, does not lock either final death component.
- When $\mathrm{emp}_t = 1$, $h_i = 0$ everywhere. UNWPP counts are baseline priors, not hard locks.

For non-HMD cells, the final admissible domain is:

$$\boxed{B_i \geq 0, \qquad 0 \leq D_i \leq P_i, \qquad M_i \leq g_i + P_i.}$$

There is no final upper bound on births and no lower bound on migration.

For HMD-locked cells:

$$(B_i^*, D_{f,i}^*, D_{m,i}^*) = (B_i^0, D_{f,i}^0, D_{m,i}^0),$$

even if the preserved death total exceeds population.

## 3.6. Exact local feasibility

Fix migration $M_i$ on a non-HMD cell and define required natural increase:

$$H_i = g_i - M_i.$$

A pair satisfying:

$$B_i \geq 0, \qquad 0 \leq D_i \leq P_i, \qquad B_i - D_i = H_i$$

exists if and only if:

$$\boxed{H_i \geq -P_i \quad \Longleftrightarrow \quad M_i \leq g_i + P_i.}$$

**Proof.** Necessity follows from $B_i - D_i \geq -P_i$. For sufficiency, if $H_i \geq 0$, choose $(B_i, D_i) = (H_i, 0)$. If $-P_i \leq H_i < 0$, choose $(B_i, D_i) = (0, -H_i)$.

Unbounded non-negative births explain the absence of a lower migration bound.

## 3.7. Exact global migration feasibility

Let:

$$\mathcal{L}_t = \{i \in \mathcal{R}_t : h_i = 1\}, \qquad \mathcal{Q}_t = \{i \in \mathcal{R}_t : h_i = 0\}.$$

Locked migration is fixed by accounting:

$$M_i^{\mathrm{HMD}} = g_i - B_i^0 + D_i^0.$$

If $\mathcal{Q}_t \neq \varnothing$, a migration vector satisfying:

$$M_i = M_i^{\mathrm{HMD}} \quad (i \in \mathcal{L}_t),$$

$$M_i \leq g_i + P_i \quad (i \in \mathcal{Q}_t), \qquad \sum_{i \in \mathcal{R}_t} M_i = 0$$

exists if and only if:

$$\boxed{\sum_{i\in\mathcal{L}_t}(g_i - B_i^0 + D_i^0) + \sum_{i\in\mathcal{Q}_t}(g_i + P_i) \geq 0.}$$

Indeed, the maximum attainable global sum is the left-hand side, while the minimum is unbounded below because at least one unlocked migration value has no lower bound.

If $\mathcal{Q}_t = \varnothing$, feasibility instead requires:

$$\sum_{i\in\mathcal{L}_t}(g_i - B_i^0 + D_i^0) = 0.$$

These are existence conditions. They do not imply that the particular three-iteration waterfill reaches a balanced vector.

# 3.8. Two-stage migration reconciliation

## Stage 1 — HMD residual override

For $h_i = 1$, set:

$$M_i^{(0)} = g_i - (B_i^0 - D_i^0).$$

For $h_i = 0$, retain:

$$M_i^{(0)} = M_i^0.$$

The HMD migration change is:

$$\Delta M_i^{(0)} = M_i^{(0)} - M_i^0.$$

Allocate that change using:

$$\eta_{f,i}^0 = \begin{cases} \dfrac{M_{f,i}^0}{M_{f,i}^0 + M_{m,i}^0}, & M_{f,i}^0 + M_{m,i}^0 \neq 0, \\ 0.488, & \text{otherwise}, \end{cases}$$

and $\eta_{m,i}^0 = 1 - \eta_{f,i}^0$.

HMD residual migration is not capped at $g_i + P_i$.

## Stage 2 — Non-HMD waterfill

For $k = 0, 1, 2$, define:

$$S^{(k)} = \sum_{i \in \mathcal{R}_t} M_i^{(k)},$$

$$\mathcal{U}_0^{(k)} = \{ i \in \mathcal{R}_t : h_i = 0,\ M_i^{(k)} < g_i + P_i \},$$

$$P_{\text{elig}}^{(k)} = \sum_{i \in \mathcal{U}_0^{(k)}} P_i.$$

Stop early if:

$$|S^{(k)}| < 0.1 \quad \text{or} \quad P_{\text{elig}}^{(k)} = 0.$$

Otherwise set:

$$\lambda^{(k)} = -\frac{S^{(k)}}{P_{\text{elig}}^{(k)}},$$

and, for eligible cells:

$$M_i^{(k+1)} = \min\left( M_i^{(k)} + P_i \lambda^{(k)},\, g_i + P_i \right).$$

All other cells remain unchanged. Each change is allocated between sexes using the current migration share, with female fallback $0.488$ when the current total is zero.

Denote the final state by $M_i^*$, and define:

$$\varepsilon_t^M = \sum_{i \in \mathcal{R}_t} M_i^*.$$

The target is $|\varepsilon_t^M| < 0.1$. A result obtained by exhausting the iteration cap must be assessed by its actual residual, not assumed to meet that tolerance.

# 3.9. Bivariate hyperbolic vital reconciliation

For $h_i = 1$, retain all preliminary vital counts. For $h_i = 0$, define:

$$H_i = g_i - M_i^*.$$

## Stage 3 — Country-bin scaling

For each bin $b$, sum over non-HMD cells in $\mathcal{R}_t$:

$$B_b^{\text{raw}} = \sum_{\substack{i \in \mathcal{R}_t \\ h_i = 0,\, b(i) = b}} B_i^0,$$

$$D_b^{\mathrm{raw}} = \sum_{\substack{i \in \mathcal{R}_t \\ h_i = 0,\, b(i) = b}} D_i^0, \qquad H_b = \sum_{\substack{i \in \mathcal{R}_t \\ h_i = 0,\, b(i) = b}} H_i.$$

If both raw totals are positive, define:

$$\Delta_b = \sqrt{H_b^2 + 4 B_b^{\mathrm{raw}} D_b^{\mathrm{raw}}},$$

$$B_b^{\mathrm{tar}} = \frac{H_b + \Delta_b}{2}, \qquad D_b^{\mathrm{tar}} = \frac{\Delta_b - H_b}{2},$$

$$s_b^B = \frac{B_b^{\mathrm{tar}}}{B_b^{\mathrm{raw}}}, \qquad s_b^D = \frac{D_b^{\mathrm{tar}}}{D_b^{\mathrm{raw}}}.$$

Otherwise:

$$s_b^B = s_b^D = 1.$$

The notation “tar” distinguishes these provisional country values from sums of final pixel outputs, which need not equal them.

## Minimum-information characterisation

For positive priors $B_0, D_0$, define the generalised KL divergence:

$$\mathcal{D}(B, D \mid B_0, D_0) = B \log \frac{B}{B_0} - B + B_0 + D \log \frac{D}{D_0} - D + D_0.$$

The strictly convex problem:

$$\underset{B>0,\, D>0}{\text{minimise}} \ \mathcal{D}(B, D \mid B_0, D_0) \quad \text{subject to} \quad B - D = H$$

has first-order conditions:

$$\log(B/B_0) + \lambda = 0, \qquad \log(D/D_0) - \lambda = 0.$$

Therefore:

$$BD = B_0 D_0.$$

Combining this with $B - D = H$ yields:

$$B = \frac{H + \sqrt{H^2 + 4 B_0 D_0}}{2}, \qquad D = \frac{\sqrt{H^2 + 4 B_0 D_0} - H}{2}.$$

Thus the positive-prior country solution is the unique Poisson minimum-information adjustment, with reciprocal scaling:

$$\boxed{s_b^B s_b^D = 1.}$$

## Stage 4 — Pixel-level closure and mortality cap

Set:

$$\overline{B}_i = s_{b(i)}^B B_i^0, \qquad \overline{D}_i = s_{b(i)}^D D_i^0.$$

The implementation applies `|| 1` to retrieved scaling factors; a factor evaluated as zero is therefore replaced by one.

For positive scaled priors, calculate:

$$\Delta_i = \sqrt{H_i^2 + 4\overline{B}_i \overline{D}_i},$$

$$(\widetilde{B}_i, \widetilde{D}_i) = \left( \frac{H_i + \Delta_i}{2}, \frac{\Delta_i - H_i}{2} \right).$$

If either scaled prior is non-positive, use the fallback:

$$(\widetilde{B}_i, \widetilde{D}_i) = ([H_i]_+, [-H_i]_+).$$

Finally:

$$\boxed{(B_i^*, D_i^*) = \begin{cases} (P_i + H_i,\ P_i), & \widetilde{D}_i > P_i, \\ (\widetilde{B}_i, \widetilde{D}_i), & \widetilde{D}_i \le P_i. \end{cases}}$$

The implementation writes $\max(0, P_i + H_i)$ in the capped branch. Under the feasibility ceiling:

$$P_i + H_i = P_i + g_i - M_i^* \ge 0,$$

so the maximum is redundant in exact arithmetic.

For positive priors, this capped solution also minimises the same KL objective subject to:

$$B - D = H_i, \qquad B \ge 0, \qquad 0 \le D \le P_i.$$

The zero-prior fallback is a separate implementation convention. It is not generally a finite-KL adjustment to the literal zero priors.

## Product invariance and the role of country scaling

For positive country totals:

$$B_b^{\mathrm{tar}} D_b^{\mathrm{tar}} = \frac{\Delta_b^2 - H_b^2}{4} = B_b^{\mathrm{raw}} D_b^{\mathrm{raw}}.$$

Hence, in exact arithmetic:

$$\overline{B}_i \overline{D}_i = s_{b(i)}^B s_{b(i)}^D B_i^0 D_i^0 = B_i^0 D_i^0.$$

Therefore:

$$\boxed{\Delta_i = \sqrt{H_i^2 + 4 B_i^0 D_i^0}}$$

on positive-prior pixels. The subsequent hyperbolic solution, including the mortality-cap decision, is unchanged by the reciprocal country scaling.

Accordingly, Stage 3 does **not** impose independent final country birth and death totals. In exact arithmetic it cancels from the pixel hyperbolic calculation.

In floating-point arithmetic, ordinary rounding can break exact product invariance. Severe subtractive cancellation can additionally round one country target to zero—deaths for large positive $H_b$, or births for large negative $H_b$—triggering the `|| 1` fallback. Overflow and underflow are further possible departures from the real-arithmetic argument.

## Exact local closure

For every feasible non-HMD cell:

$$B_i^* \geq 0, \qquad 0 \leq D_i^* \leq P_i,$$

and:

$$\boxed{B_i^* - D_i^* = H_i = g_i - M_i^*.}$$

For HMD-locked cells, closure follows directly from the residual migration override.

# 3.10. Sex-specific death and migration allocation

For $h_i = 1$:

$$(D_{f,i}^*, D_{m,i}^*) = (D_{f,i}^0, D_{m,i}^0).$$

For $h_i = 0$, define:

$$\rho_i = \begin{cases} D_{m,i}^0 / D_i^0, & D_i^0 > 0, \\ 0.512, & D_i^0 = 0. \end{cases}$$

Set:

$$D^*_{m,i} = \rho_i D^*_i, \qquad D^*_{f,i} = D^*_i - D^*_{m,i}.$$

Thus:

$$D^*_{f,i} \geq 0, \qquad D^*_{m,i} \geq 0, \qquad D^*_{f,i} + D^*_{m,i} = D^*_i.$$

There are no separately preserved single-sex death locks on non-HMD cells.

For each migration update $\Delta m_i$, use:

$$M_{f,i} \leftarrow M_{f,i} + \eta_{f,i}\Delta m_i,$$

$$M_{m,i} \leftarrow M_{m,i} + (1 - \eta_{f,i})\Delta m_i.$$

Consequently, in exact arithmetic:

$$\boxed{M^*_{f,i} + M^*_{m,i} = M^*_i.}$$

These allocations establish component-sum identities, not separate sex-specific demographic balancing equations.

# 3.11. Closure theorem and numerical implementation

## Exact-arithmetic local theorem

On $\mathcal{R}_t$, assume finite preliminary counts and:

$$M^*_i \leq g_i + P_i \quad \text{where } h_i = 0.$$

Then:

$$\boxed{g_i = B^*_i - D^*_{f,i} - D^*_{m,i} + M^*_{f,i} + M^*_{m,i}.}$$

Summing gives:

$$\boxed{\sum_{i \in \mathcal{R}_t} g_i = \sum_{i \in \mathcal{R}_t} B^*_i - \sum_{i \in \mathcal{R}_t} D^*_i + \varepsilon^M_t.}$$

Global migration balance is a separate numerical condition. If waterfill attains its target:

$$|\varepsilon^M_t| < 0.1.$$

Otherwise the actual residual remains in the aggregate identity.

## Implementation qualifications

1. **Finite waterfill.** Final balancing performs at most three updates and stops early at absolute imbalance below $0.1$, or when no eligible population remains. Saturated cells are excluded even when the requested adjustment is downward. Global feasibility alone does not guarantee algorithmic success.
2. **Newly depopulated cells.** Reconciliation skips $P_{i,t} \leq 0$, and outputs zero births, deaths, and migration there. If $P_{i,t^-} > 0$ but $P_{i,t} = 0$, then:

   $$g_{i,t} = -\frac{P_{i,t^-}}{h_t} < 0,$$

   while:

   $$B^*_{i,t} - D^*_{i,t} + M^*_{i,t} = 0.$$

   The stock loss is therefore not closed by the saved flows.
3. **Missing delta values.** The supplied audit reports that the migration summation can include positive-population cells with non-`NaN` migration even when the delta value is `NaN`, while those cells are excluded from updates. The summed domain then differs from $\mathcal{R}_t$. The clean-domain equations above require consistent finite coverage.
4. **Floating-point arithmetic.** Hyperbolic subtraction, finite optimisation, and large sums can introduce residuals before storage. Final `binary32` encoding introduces further errors. A pre-serialisation global tolerance need not remain satisfied after independently quantising all cells.
5. **No obsolete projection machinery.** This construction has no vital-event rectangle, balancing bisection, equal-weight $L_2$ birth/death projection, or threshold-based skipped migration writes of the former formulation.

For saved rasters, define the local accounting residual directly:

$$e^{\mathrm{acc}}_{i,t} = g_{i,t} - \left(B^{\mathrm{saved}}_{i,t} - D^{\mathrm{saved}}_{i,t} + M^{\mathrm{saved}}_{i,t}\right).$$

This measured residual, together with the measured saved migration sum, is the appropriate numerical diagnostic.

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# Definition 4. Compact form and simultaneous accounting constraints

## 4.1. Operators and domains

Let:

- $\mathcal{I}_{\mathrm{pix}}$ denote per-pixel elderly ceilings, coupled PAVA, secondary sex-specific PAVA, and population clamping;
- $\mathcal{I}_{\mathrm{bin}}$ denote country-bin combined-density PAVA with biological sex-ratio allocation;
- $\mathcal{W}$ denote coupled Huber–Whittaker graduation;
- $\mathcal{A}_t^{\mathrm{IPF}}$ denote three-pass redistribution followed by source compositing;
- $\mathcal{S}_t$ denote source compositing alone;
- $\mathcal{W}_t^M(\cdot;\epsilon,E)$ denote at most three upper-capped waterfill updates, with tolerance $\epsilon$ and adjustable-cell set $E$;
- $\mathcal{V}_t$ denote vital predictors, regression, calibration, source selection, and preliminary allocation;
- $\mathcal{H}(B_0,D_0,H,P)$ denote the pixel hyperbolic operator with mortality cap.

For positive priors, $\mathcal{H}$ uses:

$$\widetilde{D}=\frac{\sqrt{H^2+4B_0D_0}-H}{2},\qquad \widetilde{B}=H+\widetilde{D},$$

followed by the cap in §3.9. If either prior is non-positive, it uses $([H]_+,[-H]_+)$ before that cap.

The active population and reconciliation domains are:

$$\Omega_+(t)=\{i:P_{i,t}>0\},$$

$$\mathcal{R}_t=\{i\in\Omega_+(t):g_{i,t},M_{i,t}^0\ \text{finite}\}.$$

All flow identities refer to annualised change:

$$g_{i,t}=\frac{P_{i,t}-P_{i,t^-}}{h_t}.$$

# 4.2. Ordered construction

## A. Age–sex stocks

Construct ensemble predictions using:

$$t_{\mathrm{eff}}=\max(t,1750).$$

For $t\le 1750$:

$$\mathbf{N}_t^{\mathrm{clamp}}=\mathcal{I}_{\mathrm{pix}}(\mathbf{p}_t,\mathbf{P}_t^A),$$

$$\boxed{\mathbf{N}_t = \mathcal{S}_t(\mathbf{N}_t^{\mathrm{clamp}}; \text{sources}).}$$

For $t > 1750$, construct country populations $\mathbf{R}_t$, endpoint factors, and:

$$k_{b,t,a}^{s,(0)} = [1-\gamma(t)]k_{b,t,a}^{s,I} + \gamma(t)k_{b,t,a}^{s,W},$$

$$G_{b,t,a}^{s} = R_{b,t,a}^{s}k_{b,t,a}^{s,(0)}.$$

Then:

$$\boxed{\mathbf{N}_t = \mathcal{A}_t^{\mathrm{IPF}}(\mathbf{p}_t, \mathbf{k}_t^{(0)}, \mathbf{G}_t; \mathbf{P}_t^A, \text{sources}).}$$

Where source support permits, the target is equivalently:

$$\mathbf{G}_t = [1-\gamma(t)]\mathcal{I}_{\mathrm{bin}}(\mathbf{R}_t) + \gamma(t)\mathcal{W}(\mathbf{R}_t).$$

This equivalence requires that the scale-factor construction does not suppress positive endpoint mass on zero-support cohorts. Country targets are subsequently approximated, not exactly imposed, by the three-pass IPF procedure.

Under the stock-compatibility conditions of §1.9:

$$N_{i,t,a}^{s} \geq 0, \qquad \sum_{s,a} N_{i,t,a}^{s} = P_{i,t}.$$

Otherwise, retain the explicit stock residual:

$$e_{i,t}^{N} = \sum_{s,a} N_{i,t,a}^{s} - P_{i,t}.$$

## B. Preliminary migration and global centring

For the empirical branch, define:

$$A_{i,t}^{D,U} = \min\left(P_{i,t}, [A_{i,t}^{D_f,U}]_+ + [A_{i,t}^{D_m,U}]_+\right).$$

On $\Omega_+(t)$, construct:

$$\widetilde{M}_{i,t} = \begin{cases} \min\left(g_{i,t} - [A_{i,t}^{B,U}]_+ + A_{i,t}^{D,U}, g_{i,t} + P_{i,t}\right), & \text{empirical branch}, \\ \min\left(P_{i,t}f_{i,t}^{M}, g_{i,t} + P_{i,t}\right), & \text{modelled branch}. \end{cases}$$

Set migration to zero outside $\Omega_+(t)$. Apply:

$$\boxed{\mathbf{M}_t^0 = \mathcal{W}_t^M\left(\widetilde{\mathbf{M}}_t; \epsilon = 0.01,\ E = \Omega_+(t)\right),}$$

with population weights $\mathbf{P}_t$ and ceilings $\mathbf{g}_t + \mathbf{P}_t$.

The output satisfies the local ceiling in exact arithmetic:

$$M_{i,t}^0 \le g_{i,t} + P_{i,t}.$$

Its global residual is:

$$\varepsilon_{0,t}^M = \sum_i M_{i,t}^0.$$

The condition $|\varepsilon_{0,t}^M| < 0.01$ must be verified from the final iterate; it is not guaranteed merely by executing three updates.

Allocate migration between sexes using §2.6.

## C. Preliminary vital events and effective locks

Construct:

$$(B_{i,t}^0, D_{f,i,t}^0, D_{m,i,t}^0) = \mathcal{V}_t(\mathbf{N}, \mathbf{P}, \mathbf{X}; \text{sources}),$$

and set:

$$D_{i,t}^0 = D_{f,i,t}^0 + D_{m,i,t}^0.$$

Determine the effective whole-cell lock:

$$h_{i,t} = \mathbf{1}\left\{\text{emp}_t = 0 \ \wedge\ \left[\ell_{B,i,t} = 1 \ \vee\ (\ell_{f,i,t} = 1 \wedge \ell_{m,i,t} = 1)\right]\right\}.$$

UNWPP empirical years have $h_{i,t} = 0$. Their vital counts are reconciliation priors rather than hard constraints.

On unlocked cells, the final admissible conditions are:

$$B_{i,t} \ge 0, \qquad 0 \le D_{i,t} \le P_{i,t}, \qquad M_{i,t} \le g_{i,t} + P_{i,t}.$$

## D. Final migration and vital-event reconciliation

**Stage 1 — Replace migration on locked cells.**

Set:

$$M_{i,t}^{(0)} = \begin{cases} g_{i,t} - B_{i,t}^0 + D_{i,t}^0, & h_{i,t} = 1, \\ M_{i,t}^0, & h_{i,t} = 0. \end{cases}$$

**Stage 2 — Rebalance over unlocked cells.**

Let:

$$\mathcal{Q}_t = \{i \in \mathcal{R}_t : h_{i,t} = 0\}.$$

Apply:

$$\boxed{\mathbf{M}_t^* = \mathcal{W}_t^M \left( \mathbf{M}_t^{(0)}; \epsilon = 0.1,\ E = \mathcal{Q}_t \right).}$$

The imbalance calculation includes locked and unlocked cells in $\mathcal{R}_t$, but only unsaturated cells in $\mathcal{Q}_t$ receive adjustments. Locked migration remains unchanged.

**Stage 3 — Construct country-scaled vital priors.**

For non-HMD cells, calculate the country factors of §3.9 and set:

$$\overline{B}_{i,t} = s^B_{b(i),t} B^0_{i,t}, \qquad \overline{D}_{i,t} = s^D_{b(i),t} D^0_{i,t},$$

subject to the implementation's scaling-factor fallback.

**Stage 4 — Close local natural change.**

Define:

$$H_{i,t} = g_{i,t} - M^*_{i,t}.$$

Then:

$$\boxed{(B^*_{i,t}, D^*_{i,t}) = \begin{cases} (B^0_{i,t}, D^0_{i,t}), & h_{i,t} = 1, \\ \mathcal{H}\left(\overline{B}_{i,t}, \overline{D}_{i,t}, H_{i,t}, P_{i,t}\right), & h_{i,t} = 0. \end{cases}}$$

On positive-prior cells, reciprocal country scaling cancels from the pixel hyperbolic solution in exact arithmetic. It does not independently constrain the final country totals.

Allocate final deaths and incremental migration adjustments using §3.10, preserving:

$$D^*_{f,i,t} + D^*_{m,i,t} = D^*_{i,t},$$

$$M^*_{f,i,t} + M^*_{m,i,t} = M^*_{i,t}.$$

## 4.3. The three simultaneous accounting conditions

The system has three distinct accounting objectives:

$$\boxed{\begin{aligned} &\text{Age--sex stock reconciliation:} && \sum_{s,a} N^{s}_{i,t,a} = P_{i,t}, \\ &\text{Global migration conservation:} && \sum_{i} M^{*}_{i,t} = 0, \\ &\text{Local demographic closure:} && g_{i,t} = B^{*}_{i,t} - D^{*}_{i,t} + M^{*}_{i,t}. \end{aligned}}$$

These are not three unconditional guarantees of the implementation:

- Stock reconciliation is conditional on the selected source and clamping population.
- Global migration conservation is a finite-iteration numerical target.
- Local closure holds algebraically on the valid reconciled domain, subject to the non-HMD migration ceiling and finite arithmetic.

## Local interval accounting

On a valid reconciled cell, exact-arithmetic closure is equivalent to:

$$\boxed{P_{i,t} = P_{i,t^-} + h_t \left( B^{*}_{i,t} - D^{*}_{i,t} + M^{*}_{i,t} \right).}$$

This statement concerns total population. It does not imply cohort-by-cohort temporal consistency or separate female and male stock-flow identities.

## Aggregate interval accounting

For a complete valid domain, summing local closure gives:

$$\sum_{i} P_{i,t} - \sum_{i} P_{i,t^-} = h_t \left( \sum_{i} B^{*}_{i,t} - \sum_{i} D^{*}_{i,t} + \varepsilon^{M}_{t} \right),$$

where:

$$\varepsilon^{M}_{t} = \sum_{i} M^{*}_{i,t}.$$

If the final waterfill achieves its stopping tolerance:

$$|\varepsilon^{M}_{t}| < 0.1,$$

then the discrepancy from a births-minus-deaths-only interval identity is less than $0.1 h_t$ in exact arithmetic, provided no cells are omitted and all local identities hold.

This bound does not include serialisation errors.

## Explicit residual accounting for saved outputs

For finite saved outputs, let:

$$e_{i,t}^{\mathrm{acc}} = g_{i,t} - \left(B_{i,t}^{\mathrm{saved}} - D_{i,t}^{\mathrm{saved}} + M_{i,t}^{\mathrm{saved}}\right),$$

and:

$$\varepsilon_t^{M,\mathrm{saved}} = \sum_i M_{i,t}^{\mathrm{saved}}.$$

Then, wherever the population differences and all quantities are defined on the same complete grid:

$$\boxed{\begin{aligned}\sum_i P_{i,t} - \sum_i P_{i,t^-} = h_t \left(\sum_i B_{i,t}^{\mathrm{saved}} - \sum_i D_{i,t}^{\mathrm{saved}}\right) \\ + h_t \varepsilon_t^{M,\mathrm{saved}} + h_t \sum_i e_{i,t}^{\mathrm{acc}}.\end{aligned}}$$

This identity separates global migration imbalance from local accounting residuals.

For a newly depopulated cell whose saved flows are all zero:

$$e_{i,t}^{\mathrm{acc}} = -\frac{P_{i,t^-}}{h_t}.$$

Thus its omitted stock loss contributes $-P_{i,t^-}$ to the interval residual. Missing or non-finite inputs require separate coverage reporting rather than inclusion in a finite residual sum.

## Age–sex stock compatibility versus total-stock accounting

The principal age–sex exceptions remain:

1. unclamped WorldPop selections in 2015–2020;
2. modelled 2024–2025 cohorts clamped to the 2023 population;
3. modelled positive populations below $0.01$;
4. positive HMD cohorts passed through on zero-clamping-population cells.

These exceptions prevent an unconditional claim that all three accounting objectives hold simultaneously. They do not, by themselves, invalidate an otherwise successful flow reconciliation against the independently supplied total population $P$.

To make the distinction explicit, define aggregate age–sex population:

$$\mathcal{N}_t = \sum_i \sum_{s,a} N^s_{i,t,a},$$

and aggregate stock discrepancy:

$$E^N_t = \sum_i e^N_{i,t}.$$

Then:

$$\mathcal{N}_t = \sum_i P_{i,t} + E^N_t,$$

so:

$$\mathcal{N}_t - \mathcal{N}_{t^-} = \sum_i P_{i,t} - \sum_i P_{i,t^-} + E^N_t - E^N_{t^-}.$$

Consequently, an interval identity written using summed age–sex cohorts instead of the population-stock rasters must additionally include the change in their stock discrepancy.

## Interpretation

Under fixed inputs, parameters, processing years, and numerical execution conventions, the pipeline defines a deterministic reconstruction.

It is not a cohort-component population projection. Independently reconstructed age–sex profiles, source replacements, and the absence of explicit cohort ageing mean that a Leslie-matrix identity with migration is not imposed. Irregular time intervals alone would not prevent such a formulation; the limitation follows from the actual reconstruction operators.

The appropriate validation quantities are therefore:

$$e^N_{i,t}, \qquad e^{\mathrm{acc}}_{i,t}, \qquad \varepsilon^{M,\mathrm{saved}}_t,$$

together with source-coverage masks, lock indicators, and the set of newly depopulated cells. These distinguish structural exceptions, finite-iteration residuals, and floating-point errors.

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# Footnotes

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1. The reduced migration schedule follows the supplied report’s attribution to **Hunsinger (2022)**, adapting the Rogers–Castro migration-schedule framework of **Rogers and Castro (1981)**. Student and retirement

components are omitted in the stated implementation. The resulting schedule supplies relative sex-allocation potentials; it is not an independently estimated historical migration schedule. ↩

2. The active temporal kernel uses $t_{\text{eff}} = \max(t, 1750)$, corresponding to the stated start of HMD training coverage. It does not use a 1500 cutoff. The 80/20 same-year mixture, coverage exponent $0.3$, and 50-year temporal decay scale are modelling conventions, not accounting requirements. The 1750–1850 graduation transition and logistic steepness $8$ are likewise implementation choices; conservation arguments do not depend on their historical interpretation. ↩
3. Huber–Whittaker penalties, iteration limits, and convergence tolerances are numerical modelling choices. Biological sex-ratio envelopes are also supplied constraints, not conservation laws. Their enforcement at a graduation endpoint does not establish that every final pixel or source-composited cohort satisfies the same bounds. ↩
4. Log-tail Tukey transformations serve two separate purposes: spatial normalisation of raw OLS predictions for $M, B, D_f, D_m$, and annual migration-bound calibration from residual targets or, when those targets are absent, per-capita population growth. Vital-event target bounds instead retain their 1st/99th-percentile local/global shrinkage rule. Log-tail compression is monotone for positive scale and retains variation beyond the fences, but it does not guarantee preservation of numerical variance, country totals, or demographic accounting identities. ↩